Type MS · Two constants from factor and remainder information

Answer 1 of 13 · Question 5 part a · 5 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

5(a)	$f(-2) = 0, \quad f(-3) = 21$ $a(-2)^3 + 5b(-2)^2 + 8a(-2) - 4b = 0$ $a(-3)^3 + 5b(-3)^2 + 8a(-3) - 4b = 21$ $2b = 3a$ $41b = 51a + 21$ $a = 2^*, b = 3$	M1 A1 M1 A1cso A1 [5]
(b)	$x+2\overline{)2x^3+15x^2+16x-12}$ $\quad 2x^2+11x-6$ **ALT** $2x^3+15x^2+16x-12=(x+2)(Ax^2+Bx+C)\Rightarrow(x+2)(2x^2+11x-6)$ $2x^2+11x-6=(2x-1)(x+6)$ $(2x-1)(x+6)=0 \Rightarrow 2x-1=0, x+6=0$ $x=-6, \ -2, \ \frac{1}{2}$	M1 M1 M1 A1 [4]
		Total 9 marks

Type MS · Two constants from factor and remainder information

Answer 1 of 13 · Question 5 part a · 5 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

		Equates coefficients to find the 3TQ factor $2x^3+15x^2+16x-12=(x+2)(Ax^2+Bx+C)\Rightarrow(x+2)(2x^2+11x-6)$ Must get as far as $(x+2)(2x^2+\cdots)$ for the mark.
	M1	For attempting to factorise **their** 3TQ, but it must be a 3TQ $2x^2+11x-6=(2x-1)(x+6)$ Refer to general guidance for what constitutes an attempt to factorise.
	M1	An attempt to solve f $(x) = 0$
	A1	For $x=-6,\ -2,\ \frac{1}{2}$ **Note:** Correct answers with no working scores M0M0M0A0

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Two constants from factor and remainder information

Answer 1 of 13 · Question 5 part a · 5 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

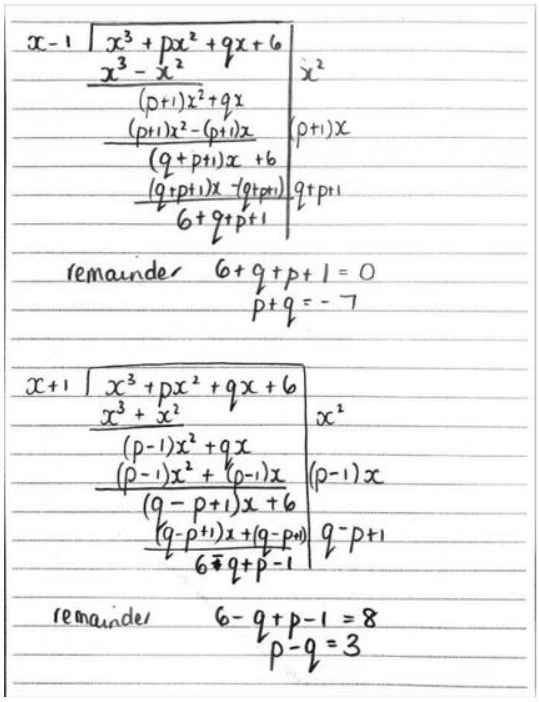
Question number	Scheme	Marks

Type MS · Two constants from factor and remainder information

Answer 2 of 13 · Question 4 part a.i · 0 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

4 a Mark parts (i) and (ii) together	$(f(1)=)1^3+p\times1^2+q\times1+6=0$ oe	M1
	or	
	$(f(-1)=)(-1)^3+p\times(-1)^2+q\times(-1)+6=8$ oe	
	$p+q=-7$ oe	A1
	$p-q=3$ oe	A1
	Solving simultaneously $2p=-4$ or $2q=-10$	M1
	$p=-2$ *	A1 cso
	and $q=-5$	B1 (A1 on ePen) cso
		(6)
b	$(f(x)=x^3-2x^2-5x+6=0)$	
	$(x-1)(x^2-x-6)=0$	M1
	$[(x-1)](x+2)(x-3)=0$	M1
	$x=1,-2,3$	A1
		(3)
		Total 9 marks



Type MS · Two constants from factor and remainder information

Answer 2 of 13 · Question 4 part a.i · 0 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For a fully correct substitution of either 1 into f(x) = 0 **OR** -1 into f(x) = 8 as shown. The bracketing for the substitution of -1 must be correct, but may be recovered later. A correct equation can imply this mark.
	A1	For a fully correct substitution of 1 into f(x) = 0 **AND** -1 into f(x) = 8 as shown. The bracketing for the substitution of -1 must be correct, but may be recovered later. A correct equation can imply this mark.
	A1	For $p+q=-7$ **and** $p-q=3$ oe The equations do not need to be simplified.
	Note, if students choose to do long division, to be comparable with the main scheme, this must be fully correct and complete, to allow them to arrive at the correct equations to be awarded the method mark. See the example included. The accuracy marks can be then awarded as stated above. There is an example under the main mark scheme. **If in any doubt, please send to review.**	
	M1	For an attempt to solve their equations simultaneously, allow one processing error. Must eliminate one variable and arrive at a value for p or q.
	A1 cso	For $p=-2$
	B1 (A1 on ePen)	$q=-5$ Note this is an independent accuracy mark, allowing students to use the given value of p to find q.
(b)	M1	For $(x-1)(x^2 \pm Ax-6)=0 \qquad A\neq 0$ This mark can be awarded for sight of $x^2 \pm Ax-6 \qquad A\neq 0$
	M1	For $[(x-1)](x+2)(x-3)=0$ **For this mark, for the correct or ft their quadratic:** Allow any minimally acceptable attempt to solve the quadratic, by factorising, completing the square or use of the quadratic formula – see general guidance, leading to two values of x in addition to $x = 1$. It is not necessary to see = 0.
	A1	For $x=1,-2,3$ If 3 three correct solutions appear with no working M1 M1 A1 as the question has not stated show working or solve algebraically.

Type MS · Two constants from factor and remainder information

Answer 2 of 13 · Question 4 part a.i · 0 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

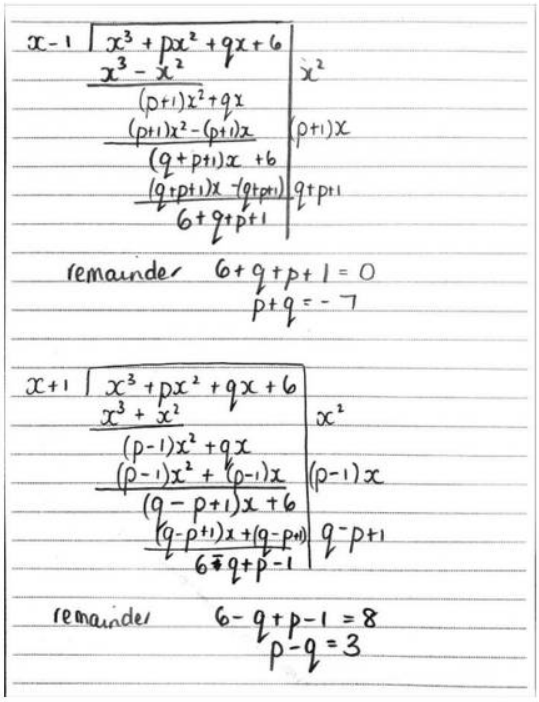
Question number	Scheme	Marks
	$kx^2+(k+1)x-1 \quad k$	

Type MS · Two constants from factor and remainder information

Answer 3 of 13 · Question 4 part a.ii · 6 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

4 a Mark parts (i) and (ii) together	$(f(1)=)1^3+p\times1^2+q\times1+6=0$ oe	M1
	or	
	$(f(-1)=)(-1)^3+p\times(-1)^2+q\times(-1)+6=8$ oe	
	$p+q=-7$ oe	A1
	$p-q=3$ oe	A1
	Solving simultaneously $2p=-4$ or $2q=-10$	M1
	$p=-2$ *	A1 cso
	and $q=-5$	B1 (A1 on ePen) cso
		(6)
b	$(f(x)=x^3-2x^2-5x+6=0)$	
	$(x-1)(x^2-x-6)=0$	M1
	$[(x-1)](x+2)(x-3)=0$	M1
	$x=1,-2,3$	A1
		(3)
		Total 9 marks



Type MS · Two constants from factor and remainder information

Answer 3 of 13 · Question 4 part a.ii · 6 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For a fully correct substitution of either 1 into f(x) = 0 **OR** –1 into f(x) = 8 as shown. The bracketing for the substitution of –1 must be correct, but may be recovered later. A correct equation can imply this mark.
	A1	For a fully correct substitution of 1 into f(x) = 0 **AND** –1 into f(x) = 8 as shown. The bracketing for the substitution of –1 must be correct, but may be recovered later. A correct equation can imply this mark.
	A1	For $p+q=-7$ **and** $p-q=3$ oe The equations do not need to be simplified.
	Note, if students choose to do long division, to be comparable with the main scheme, this must be fully correct and complete, to allow them to arrive at the correct equations to be awarded the method mark. See the example included. The accuracy marks can be then awarded as stated above. There is an example under the main mark scheme. **If in any doubt, please send to review.**	
	M1	For an attempt to solve their equations simultaneously, allow one processing error. Must eliminate one variable and arrive at a value for p or q.
	A1 cso	For $p=-2$
	B1 (A1 on ePen)	$q=-5$ Note this is an independent accuracy mark, allowing students to use the given value of p to find q.
(b)	M1	For $(x-1)(x^2 \pm Ax-6)=0 \qquad A\neq 0$ This mark can be awarded for sight of $x^2 \pm Ax-6 \qquad A\neq 0$
	M1	For $[(x-1)](x+2)(x-3)=0$ **For this mark, for the correct or ft their quadratic:** Allow any minimally acceptable attempt to solve the quadratic, by factorising, completing the square or use of the quadratic formula – see general guidance, leading to two values of x in addition to x = 1. It is not necessary to see = 0.
	A1	For $x=1,-2,3$ If 3 three correct solutions appear with no working M1 M1 A1 as the question has not stated show working or solve algebraically.

Type MS · Two constants from factor and remainder information

Answer 3 of 13 · Question 4 part a.ii · 6 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks
	$\left(\quad k \quad\right) \quad \left(\quad k \quad\right) \quad kx^2+(k+1)x-1 \quad k$	

Type MS · Two constants from factor and remainder information

Answer 4 of 13 · Question 2 part a · 6 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

2 (a)	$f(1)=1+p+q=-12$	M1
	$p+q=-13$	A1
	$f(4)=64+4p+q=30$	M1
	$4p+q=-34$	A1
	$3p=-21$	M1
	$p=-7$ and $q=-6$	A1 (6)
(b)	$3^3-7\times3-6=27-21-6=0$ *	B1 cso (1)
(c)	$(x-3)(x^2+3x+2)$	M1
	$(x-3)(x+2)(x+1)$	M1 A1 (3)
(d)	$x=3$ $x=-2$ $x=-1$	B1 ft (1)
		[11]

	Notes
(a)	
M1	For substitution of 1 into f(x)
A1	For $p+q=-13$ oe
M1	For substitution of 4 into f(x)
A1	For $4p+q=-34$ oe
M1	For solving simultaneously
A1	$p=-7$ and $q=-6$
(b)	
B1cso	For substituting 3 into f(x) and obtaining the given result
(c)	
M1	For $(x-3)(x^2+3x+2)$
M1	For factorising the quadratic
A1	$(x-3)(x+2)(x+1)$
(d)	
B1ft	For $x=3$ $x=-2$ $x=-1$ or follow through part (c)

Type MS · Two constants from factor and remainder information

Answer 4 of 13 · Question 2 part a · 6 marks

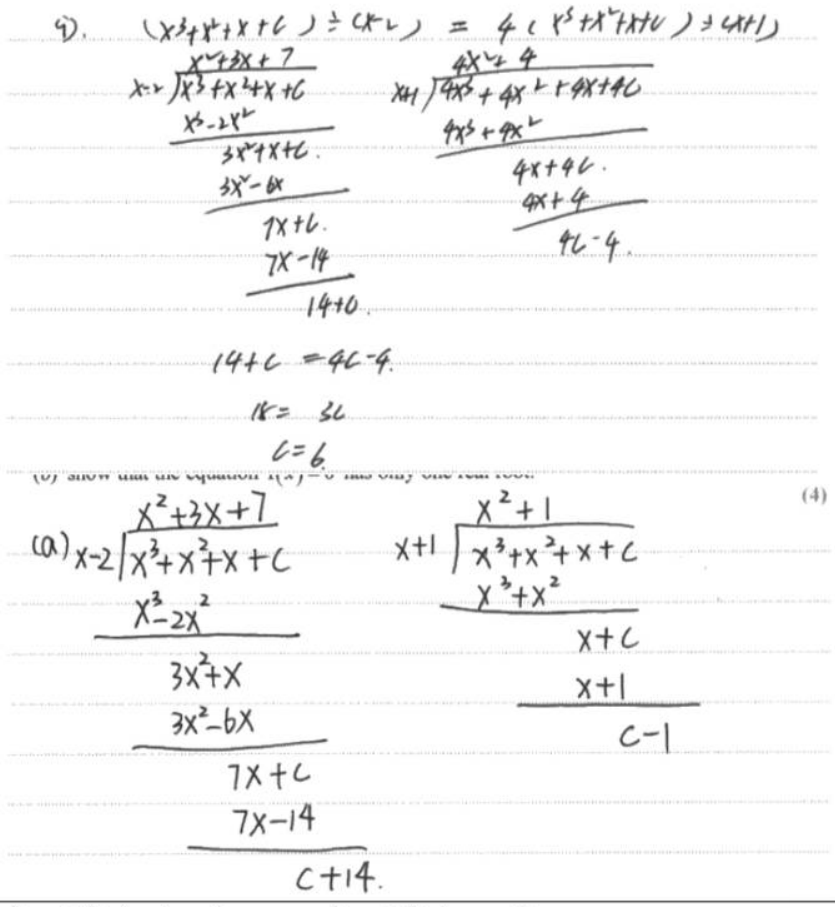
2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Two constants from factor and remainder information

Answer 5 of 13 · Question 5 part a · 4 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

5 (a)	$4[(-1)^3+1^2+(-1)+c]=2^3+2^2+2+c$ $-4+4c=14+c$ oe $c=6$ *	M1 M1 ddM1 A1*cso [4]
		
(b)	$[(x+2)](x^2+Ax+3)$ or $[(x+2)](x^2-x+B)$ $[(x+2)](x^2-x+3)$ $(b^2-4ac=)(-1)^2-4(1)(3)<0$ therefore only one (real) root * or $\left(x-\frac{1}{2}\right)^2=-\frac{11}{4}$ and $-\frac{11}{4}<0$ therefore only one (real) root	M1 A1 M1 A1*cso [4]
		Total 8 marks

Type MS · Two constants from factor and remainder information

Answer 5 of 13 · Question 5 part a · 4 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correct substitution of – 1 **or** 2 into f(x)
	M1	For correct substitution of – 1 **and** 2 into f(x)
	For the first two marks: Where candidates employ simultaneous equations, mark the work for the substitution Accept eg -1 for $(-1)^3$ 4 for 2^2 Allow – 1 + c **with no errors** to imply substitution of – 1 Allow 14 + c **with no errors** to imply substitution of 2	
	ddM1	For $4[(-1)^3+1^2+(-1)+c]=2^3+2^2+2+c$ or condone the 4 on the 'wrong side'. $(-1)^3+1^2+(-1)+c=4[2^3+2^2+2+c]$ Dependent on both previous method marks. For forming simultaneous equations eg $4R = 14 + c$ $R = -1 + c$ Condone $R = 14 + c$ $4R = -1 + c$ The work doesn't have to be simplified for this mark.
	A1*cso	Correct solution, no errors or omissions, minimum steps as shown. The steps will need to be checked. Note there are different notations for long division.
	For a route that uses long division – examples in mark scheme. M1 For completing a correct long division, giving a correct remainder M1 for completing 2 correct long divisions, giving both correct remainders ddM1 for the correct equation or condone the 4 on the wrong side. A1 Correct solution, no errors or omissions, minimum steps as shown. The steps will need to be checked.	
(b)	M1	For $[(x+2)](x^2+Ax+3)$ or $[(x+2)](x^2-x+B)$ **using any method** – there must be a quadratic to award this mark. Mark the quadratic seen, not the method.
	A1	For x^2-x+3
	M1	For calculating the discriminant correctly for their 3 term quadratic. If working with the correct quadratic – 11 will imply this mark. If not, they must show at least partial substitution (their b^2 and $4ac$ evaluated will be fine) And condone/allow $(1)^2-4(1)(3)<0$ since $(-1)^2=(1)^2$ Or full correct completion of square and $-\frac{11}{4}<0$ oe. The discriminant may be calculated in the quadratic formula.
	A1*cso	Fully correct solution and short conclusion State < 0 and (can be "no (real) roots", (x =) "– 2 is the only (real) root", "shown" or #), no errors. Minimum steps as shown.
	Send any work on complex numbers to Review.	

Type MS · Two constants from factor and remainder information

Answer 5 of 13 · Question 5 part a · 4 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Two constants from factor and remainder information

Answer 6 of 13 · Question 4 part a · 6 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

Number		
4 (a)	$f(2)=2\times2^3+a\times2^2+b\times2+18=0$	M1
	$f'(x)=6x^2+2ax+b \Rightarrow f'(2)=6\times2^2+2\times a\times2+b=5$	M1M1
	$4a+2b+34=0$	
	$4a+b+19=0$	A1
	$\Rightarrow b=-15,\ a=-1$	M1A1 [6]
(b)	$x-2\overline{)2x^3-x^2-15x+18}$ $\quad 2x^2+3x-9$	M1
	$2x^2+3x-9=(x+3)(2x-3)$	
	$\Rightarrow (x-2)(x+3)(2x-3)$	M1A1 [3]
(c)	$x=2,\ -3,\ \frac{3}{2}$	B2ft [2]
		Total 11 marks

Type MS · Two constants from factor and remainder information

Answer 6 of 13 · Question 4 part a · 6 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks
3(a)	$\frac{dy}{dx} = 2e^{2x}\left(x^2+1\right)+e^{2x}\left(2x\right)$	M1A1A1 [3]
(b)	When $x=0$ $\frac{dy}{dx} = 2\times1\times1+1\times0=2$ $\quad y=e^{2\times0}\left(0+1\right)=1$ $y-1=2\left(x-0\right)\Rightarrow y=2x+1$	B1B1 B1 [3]
		Total 6 marks

(a)	
M1	Attempted use of the product rule. Sum of two terms (either way round) with $x^n \to x^{n-1}$ (Condone e^{2x} instead of $2e^{2x}$) Once the correct answer is seen ISW. This mark may be implied by the sum of two terms with one of the two terms correct.
A1	Either term correct
A1	Both terms correct
(b)	
B1	When $x=0$ $\frac{dy}{dx}=2$
B1	When $x=0$ $y=1$
B1	$y=2x+1$

Type MS · Two constants from factor and remainder information

Answer 6 of 13 · Question 4 part a · 6 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks
4 (a)	$f(2)=2\times2^3+a\times2^2+b\times2+18=0$ $f'(x)=6x^2+2ax+b \Rightarrow f'(2)=6\times2^2+2\times a\times2+b=5$ $4a+2b+34=0$ $4a+b+19=0$ $\Rightarrow b=-15,\ a=-1$	M1 M1M1 A1 M1A1 [6]
(b)	$\begin{array}{r} 2x^2+3x-9 \\ x-2\overline{)2x^3-x^2-15x+18} \end{array}$ $2x^2+3x-9=(x+3)(2x-3)$ $\Rightarrow (x-2)(x+3)(2x-3)$	M1 M1A1 [3]
(c)	$x=2,\ -3,\ \frac{3}{2}$	B2ft [2]
		Total 11 marks
(a)		
M1	$f(2)=0$ leading to an equation in a and b	
M1	Attempt to differentiate	
M1	$f'(2)=5$ leading to an equation in a and b	
A1	$4a+2b+34=0$ and $4a+b+19=0$	
M1	Solving simultaneously	
A1	$b=-15,\ a=-1$	
(b)		
M1	Dividing by $x-2$ to obtain a 3TQ	
M1	Factorising the 3TQ	
A1	All 3 terms correct	
(c)		
B2 ft	$x=2,\ -3,\ \frac{3}{2}$ (**B1** for 2 correct)	

Type MS · Two constants from factor and remainder information

Answer 6 of 13 · Question 4 part a · 6 marks

2020 JAN · Paper 01 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Two constants from factor and remainder information

Answer 7 of 13 · Question 5 part a · 5 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks
5(a)	$f(1)=2\times1^3+a\times1^2-14\times1+b=0 \quad \Rightarrow \quad a+b=12$ $f(4)=2\times4^3+a\times4^2-14\times4+b=39 \Rightarrow 16a+b=-33$ $\Rightarrow 15a=-45 \Rightarrow a=-3^*$ $\Rightarrow b=15$	M1 M1 M1A1 cso B1 [5]
	ALT – polynomial division	
	$(2x^3+ax^2-14x+b)\div(x-1)=2x^2+(a+2)x-14+a+2$ and comparison of final step of division with b to obtain an equation **or** $(2x^3+ax^2-14x+b)\div(x-4)=2x^2+(a+8)x+(18+4a)$ r 39 and comparison of final step of division with obtaining remainder 39 to obtain an equation $(2x^3+ax^2-14x+b)\div(x-1)=2x^2+(a+2)x-14+a+2$ and comparison of final step of division with b to obtain an equation **and** $(2x^3+ax^2-14x+b)\div(x-4)=2x^2+(a+8)x+(18+4a)$ r 39 and comparison of final step of division with obtaining remainder 39 to obtain an equation $b=12-a$ $b+4(18+4a)=39$ $a=-3^*$ $b=15$	[M1 M1 M1 A1cso B1]
(b)	$(x-1)\overline{)2x^3-3x^2-14x+15}$ $\quad 2x^2-x-15$ $f(x)=(x-1)(2x+5)(x-3)$	M1A1 dM1A1 [4]
(c)	$p=-\frac{5}{2} \quad q=1 \quad r=3$	B1ft B1ft B1ft [3]

Type MS · Two constants from factor and remainder information

Answer 7 of 13 · Question 5 part a · 5 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 5 · find part a in this mark scheme extract

(d)	$A = \left\lvert \int_1^3 \left(2x^3 - 3x^2 - 14x + 15\right) \mathrm{d}x \right\rvert + \int_{-\frac{5}{2}}^{1} \left(2x^3 - 3x^2 - 14x + 15\right) \mathrm{d}x$	M1
	$A = \left\lvert \left[\frac{2x^4}{4} - \frac{3x^3}{3} - \frac{14x^2}{2} + 15x\right]_1^3 \right\rvert + \left[\frac{2x^4}{4} - \frac{3x^3}{3} - \frac{14x^2}{2} + 15x\right]_{-\frac{5}{2}}^{1}$	M1*
	$A = \left\lvert \left[\left(\frac{3^4}{2} - 3^3 - 7\times 3^2 + 15\times 3\right) - \left(\frac{1^4}{2} - 1^3 - 7\times 1^2 + 15\times 1\right)\right] \right.$ $\left. + \left[\left(\frac{1^4}{2} - 1^3 - 7\times 1^2 + 15\times 1\right) - \left(\frac{\left(-\frac{5}{2}\right)^4}{2} - \left(-\frac{5}{2}\right)^3 - 7\times\left(-\frac{5}{2}\right)^2 + 15\times\left(-\frac{5}{2}\right)\right)\right] \right\rvert$	dM1
	$A = 12 + \frac{1715}{32} = \frac{2099}{32}$ or 65.59375	A1 [4]
		Total 16 marks

Type MS · Two constants from factor and remainder information

Answer 7 of 13 · Question 5 part a · 5 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 5 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For use of f (1) = 0 **or** use of f (4) = 39
	M1	For use of f (1) = 0 **and** use of f (4) = 39
	M1	For solving the simultaneous equations in a and b Must see at least one line of working before a or b obtained. Must see an equation in just one variable.
	A1cso	For showing $a = -3$* Must see expression only in terms of a
	B1	For the correct value of $b = 15$ seen. Note: this is an A mark in epen.
	ALT – polynomial division	
	M1	Correct method for forming one equation by polynomial division together with comparison of the final step of the division with the required result (no remainder for division by $(x-1)$ and remainder 39 for division by $(x-4)$)
	M1	Correct method for forming **both** equations by polynomial division together with comparison of the final step of the division with the required result (no remainder for division by $(x-1)$ and remainder 39 for division by $(x-4)$)
	M1	For solving the simultaneous equations in a and b Must see at least one line of working before a or b obtained.
	A1cso	For showing $a = -3$* Must see expression only in terms of a ie $ka = -3k$ before $a = -3$
	B1	For the correct value of $b = 15$ seen. Note: this is an A mark in epen.
	Assuming $a = -3$ and solving to find $b = 15$ scores maximum M1M0M0A0A1	
(b)	**Condone (b) completed in (c)**	
	M1	For carrying out polynomial division/comparing coefficients to find the quadratic factor. Must reach $2x^2 \pm \cdots$. Allow statement of quadratic factor if polynomial division by $(x-1)$ used in (a).
	A1	For the correct quadratic factor.
	dM1	For factorising their 3TQ See general guidance for what constitutes an attempt to factorise a 3TQ.
	A1	For the correct factorisation of f (x) on one line. Condone omission of f$(x) =$
	Note: complete factorisation seen with no method scores M0A0M0A0.	
(c)	B1ft	For one correct value for any of p, r or q FT their linear factors in (b). Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.
	B1ft	For two correct values. FT their linear factors in (b). Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.
	B1ft	For all three correct values. FT their linear factors in (b), but **1 must be one of the values**. Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.

Type MS · Two constants from factor and remainder information

Answer 7 of 13 · Question 5 part a · 5 marks

2023 JUN · Paper 01 (regional R) · extract 4 of 5 · find part a in this mark scheme extract

(d)	M1	For a correct statement for the area of the shaded regions with their limits correctly used. Minimum required is: $\int_p^q (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x - \int_q^r (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x$ with $p < q < r$ shown as numerical limits or $\int_p^q (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x + \left\lvert \int_q^r (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x \right\rvert$ with $p < q < r$ shown as numerical limits
	M1*	For an attempt to integrate their expression. Minimum required is: • two powers of terms to increase by 1 • no powers of terms to decrease Condone $+c$
	dM1	For substituting their limits correctly to evaluate the integrated expression. Must show the substitution if their integral or their limits are incorrect. Correct exact answer following correct integral with correct limits implies this mark. Condone $+c$ as long as this is cancelled out during the substitution.
	A1	For the correct area. Exact area required.

Type MS · Two constants from factor and remainder information

Answer 7 of 13 · Question 5 part a · 5 marks

2023 JUN · Paper 01 (regional R) · extract 5 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Two constants from factor and remainder information

Answer 8 of 13 · Question 4 part a.i · 0 marks

2024 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Two constants from factor and remainder information

Answer 9 of 13 · Question 4 part a.ii · 6 marks

2024 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Two constants from factor and remainder information

Answer 10 of 13 · Question 7 part a · 5 marks

2025 NOV · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

7(a)	$\left[\mathrm{f}\left(-\frac{1}{2}\right)=\right]2\times\left(-\frac{1}{2}\right)^3+A\times\left(-\frac{1}{2}\right)^2+B\times\left(-\frac{1}{2}\right)-20=0\left[\Rightarrow A-2B=81\right]$	M1
	$\left[\mathrm{f}'(x)=\right]6x^2+2Ax+B$	M1
	$\left[\mathrm{f}'(1)=\right]6\times1^2+2A\times1+B=-27\left[\Rightarrow 2A+B=-33\right]$	M1
	$A-2B=81$ oe	
	$2A+B=-33$ oe	dddM1
	$\Rightarrow A=3^*,\quad B=-39$	A1 cso [5]
(b)	$2x+1\overline{)2x^3+3x^2-39x+20}$ $\quad x^2+x-20$	M1
	$x^2+x-20=(x-4)(x+5)$	dM1
	$\Rightarrow\left[x=\right]-\frac{1}{2},4,-5$	A1 [3]
(c)	$\left[A=\right]\int_{"-5"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x+\left\lvert\int_{"-\frac{1}{2}"}^{"4"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x\right\rvert$	M1
	oe eg $\int_{"-5"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x+\int_{"4"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x$	
	$\left[A=\right]\left[\frac{2x^4}{4}+\frac{3x^3}{3}-\frac{39x^2}{2}-20x\right]_{[-5]}^{\left[-\frac{1}{2}\right]}+\left[\frac{2x^4}{4}+\frac{3x^3}{3}-\frac{39x^2}{2}-20x\right]_{\left[-\frac{1}{2}\right]}^{[4]}$	M1
	$\left[A=\right]\left\{\left[\frac{2\left(-\frac{1}{2}\right)^4}{4}+\frac{3\left(-\frac{1}{2}\right)^3}{3}-\frac{39\left(-\frac{1}{2}\right)^2}{2}-20\left(-\frac{1}{2}\right)\right]-\left[\frac{2(-5)^4}{4}+\frac{3(-5)^3}{3}-\frac{39(-5)^2}{2}-20(-5)\right]\right\}$	
	$\left[+\right]\left\{\left[\frac{2(4)^4}{4}+\frac{3(4)^3}{3}-\frac{39(4)^2}{2}-20(4)\right]-\left[\frac{2\left(-\frac{1}{2}\right)^4}{4}+\frac{3\left(-\frac{1}{2}\right)^3}{3}-\frac{39\left(-\frac{1}{2}\right)^2}{2}-20\left(-\frac{1}{2}\right)\right]\right\}$ oe	M1
	$\left[A=\right]\frac{6561}{16}$ or 410.0625 oe	A1 [4]
		Total 12 marks

Type MS · Two constants from factor and remainder information

Answer 10 of 13 · Question 7 part a · 5 marks

2025 NOV · Paper 02 · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	Substitutes $-\frac{1}{2}$ into f(x) and sets = 0 Accept as a minimum $2\times\left(-\frac{1}{8}\right)+A\times\frac{1}{4}+B\times\left(-\frac{1}{2}\right)-20=0$ or $-\frac{1}{4}+A\times\frac{1}{4}+B\times\left(-\frac{1}{2}\right)-20=0$
	M1	For correctly differentiating f(x)
	M1	Substitutes 1 into their derivative, and sets = – 27 Accept $2A+B=-33$ with no errors. This isn't a dependent mark, but candidates must be substituting into a changed expression with at least 2 terms in x
	dddM1	Solves their linear simultaneous equations by any method, allow one error in solving. Minimal working here is fine, but there must be **some** working. Candidates who show no working to solve the simultaneous equations will not gain this mark (and therefore not the next mark) As this is a 'show' question, correct answers cannot imply this method mark. Dependent on previous 3 method marks.
	A1*cso	For the correct value of A [which is given in the question] and the correct value of B. Minimum steps as shown/stated, no errors or omissions.
(b)	M1	Award the mark if the candidate is seen to be attempting any method, such as algebraic division or equating coefficients (note this will often have minimal working) and attains: $x^2\pm Cx-20$ or $x^2+x\pm D$ $C,D\neq 0$ It is not sufficient to show simply a product of linear factors.
	dM1	For correctly factorising their quadratic factor or correctly using any method to solve their quadratic factor = 0 Dependent on previous method mark.
	A1	For all three roots. Candidates are told to use algebra, so correct answers alone cannot imply M marks for this question. This means correct answers alone will be 0 marks overall.
The majority of candidates are completing this question, using the factor of $(2x + 1)$ given. Occasionally, we are seeing a re-start using a different factor – probably deduced from a calculator, but if sufficient algebra is shown, we will award the marks. For $(x + 5)$ need to see $2x^2-7x-4$ with $2x^2$ correct and either the – 7 or – 4 correct For $(x – 4)$ need to see $2x^2+11x+5$ with $2x^2$ correct and either the +11 or +5 correct for the first M mark – the next 2 marks then follow.		

Type MS · Two constants from factor and remainder information

Answer 10 of 13 · Question 7 part a · 5 marks

2025 NOV · Paper 02 · extract 3 of 4 · find part a in this mark scheme extract

(c)	M1	For a correct statement of the required area. Allow use of their limits from part (b), even if working wasn't shown. However, no limit may be 0 This mark may be implicit in further working and may even be awarded at the stage when a negative numerical area has been made positive.
	M1	For a minimally acceptable attempt at integration of an expression of the form $2x^3+3x^2+Px-20$ $P\neq 0$. The modulus sign and the addition between the 2 calculations do not need to be seen for this mark. See general guidance, additionally, no power of x to decrease. Limits do no need to be present for this mark.
	M1	For a full and correct substitution of each of their limits (their roots from part (b), no limit 0) into any changed expression with a minimum of 2 terms. The modulus sign and the addition between the 2 calculations do not need to be seen for this mark. A correct final answer, if the integration is shown, may imply this mark.
	A1	For the correct exact area. The correct final answer may imply the first and third method marks, but may not imply the second method mark (ie must show integration). If no integration is shown, with a correct answer, the only possible marks are M1 M0 M1 A0 (if it's clear somewhere in working that the candidate has split the area). or M0 M0 M1 A0 (if the answer is correct with no reference to splitting the area).
If candidates work with $\int_{"-5"}^{"4"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x$ they **may** attain up to M0 M1 M1 A0		

Type MS · Two constants from factor and remainder information

Answer 10 of 13 · Question 7 part a · 5 marks

2025 NOV · Paper 02 · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Two constants from factor and remainder information

Answer 11 of 13 · Question 11 part a · 6 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

11 (a)	$f(x)=\int ax^2-14x-10\,dx=\frac{ax^3}{3}-\frac{14x^2}{2}-10x+c$	M1A1
	$f(4)=\frac{a\times4^3}{3}-\frac{14\times4^2}{2}-10\times4+c=0\Rightarrow\frac{64a}{3}-152+c=0$	M1
	$f(-1)=\frac{a\times(-1)^3}{3}-\frac{14\times(-1)^2}{2}-10\times(-1)+c=25\Rightarrow-\frac{a}{3}-22+c=0$	M1
	$\frac{64a}{3}-152+c=0$ $-\frac{a}{3}-22+c=0$	
	$\Rightarrow 152-\frac{64a}{3}=22+\frac{a}{3}\Rightarrow a=6$	M1A1 [6]
(b)	$c=22+\frac{6}{3}=24$	B1
	$f(x)=2x^3-7x^2-10x+24$	
	$x-4\overline{)2x^3-7x^2-10x+24}$ $\quad 2x^2+x-6$	M1A1
	$f(x)=(x-4)(2x-3)(x+2)=0$	dM1A1
	$x=4,\ \frac{3}{2},\ -2$	A1 [6]
	ALT $2x^3-7x^2-10x+24=(x-4)(ax^2+bx+c)$ $a=2, b=1, c=-6$	[M1
	$f(x)=(x-4)(2x^2+x-6)$	A1
	$f(x)=(x-4)(2x-3)(x+2)=0$	dM1A1
	$x=4,\frac{3}{2},-2$	A1]
		Total 12 marks

Type MS · Two constants from factor and remainder information

Answer 11 of 13 · Question 11 part a · 6 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

	M1	For correct method to solve the equations simultaneously to find a A correct intermediate step is required e.g. $65a = 390$
	A1 cso	For fully correct working leading to $a = 6$
(b)	**B1**	For correctly identifying $c = 24$
	M1	For attempt to divide f(x) by $x - 4$ Must get as far as $2x^2 + \cdots$
	A1	For correct division of f(x) by $x - 4$ to obtain $2x^2 + x - 6$
	dM1	For an attempt to factorise their 3TQ which must come from a cubic. See General Guidance on what constitutes an attempt to factorise.
	A1	For obtaining correct factorisation of the cubic: $(x - 4)(2x - 3)(x + 2)$
	A1	For all three correct solutions of the equation: $x = 4, \frac{3}{2}, -2$
Alternative method		
	M1	For an attempt to find the quadratic factor that multiplies $x - 4$ to give f(x) Must get as far as $f(x) = (x - 4)(2x^2 + bx + c)$
	A1	For correctly comparing coefficients to obtain $2x^2 + x - 6$
	dM1	For an attempt to factorise their 3TQ which must come from a cubic. See General Guidance on what constitutes an attempt to factorise.
	A1	For obtaining correct factorisation of the cubic: $(x - 4)(2x - 3)(x + 2)$
	A1	For all three correct solutions of the equation: $x = 4, \frac{3}{2}, -2$
	Note: Correct solution seen with no working scores B0M0A0M0A0A0	

Type MS · Two constants from factor and remainder information

Answer 11 of 13 · Question 11 part a · 6 marks

2021 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

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Type MS · Two constants from factor and remainder information

Answer 12 of 13 · Question 4 part a · 5 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

4 (a)	f (±1)=0 **or** f (±2)= –5	M1
	$-1 + 1p + -1q + 7 = 0$ $(p - q + 6 = 0)$ **and** $-8 + 4p + -2q + 7 = -5$ $(4p - 2q - 1 = -5)$	A1
	$4(q - 6) - 2q = -4$ $(2q = 20)$	dM1
	$p = 4$ $q = 10$	A1 A1 [5]
	ALT – polynomial division $(x^3 + px^2 + qx + 7) \div (x + 1) = x^2 + (p - 1)x + q - p + 1$ and comparison of final step of division with 7 to obtain an equation **or** $(x^3 + px^2 + qx + 7) \div (x + 2) =$ $x^2 + (p - 2)x + (q - 2p + 4)$ remainder -5 and comparison of final step of division with obtaining remainder -5 to obtain an equation	M1
	$(x^3 + px^2 + qx + 7) \div (x + 1) = x^2 + (p - 1)x + q - p + 1$ and comparison of final step of division with 7 to identify $q - p + 1 = 7$ **and** $(x^3 + px^2 + qx + 7) \div (x + 2) =$ $x^2 + (p - 2)x + (q - 2p + 4)$ remainder -5 and comparison of final step of division with obtaining remainder -5 to identify $7 - 2(q - 2p + 4) = -5$	A1
	$(p + 6) - 2p = 2$ $(-p = -4)$	dM1
	$p = 4$ $q = 10$	A1 A1
(b)	$x+1\overline{)x^3 + "4"x^2 + "10"x + 7}$ $(+3x + 7)$ x^2 $x^3 + x^2$ $"3"x^2$ or $x^3 + 4x^2 + 10x + 7 \equiv (x + 1)(x^2 + Ax + B)$	M1
	"3"^{2} – 4(1)("7") = a value	dM1
	–19 and a conclusion drawn e.g. the discriminant is negative so only one real root	A1 [3]
	ALT – use of completing the square $x+1\overline{)x^3 + "4"x^2 + "10"x + 7}$ $(+3x + 7)$ x^2 $x^3 + x^2$ $"3"x^2$ or $x^3 + 4x^2 + 10x + 7 \equiv (x + 1)(x^2 + Ax + B)$	M1

Type MS · Two constants from factor and remainder information

Answer 12 of 13 · Question 4 part a · 5 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

	$\left(x+\frac{3}{2}\right)^2+\frac{19}{4}>0$	dM1
	$\left(x+\frac{3}{2}\right)^2+\frac{19}{4}>0$ and a conclusion drawn e.g. the completed square form is always greater than 0 so only one real root	A1
		Total 8 marks

Type MS · Two constants from factor and remainder information

Answer 12 of 13 · Question 4 part a · 5 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

	dM1	Attempt at completing the square See general guidance for what constitutes an attempt at completing the square
	A1	Correct completed square form and draws a conclusion. There must be some conclusion drawn, but it can be as simple as writing '#' or "shown".

Type MS · Two constants from factor and remainder information

Answer 12 of 13 · Question 4 part a · 5 marks

2022 JUN · Paper 02 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Two constants from factor and remainder information

Answer 13 of 13 · Question 5 part a · 5 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
5(a)	$f(-2)=-16a+4+2b+3a=0$ and $f(1)=2a+1-b+3a=0$ $-13a+4+2b=0$ and $5a+1-b=0 \Rightarrow a=2^*$ $b=11$	M1M1 M1A1cso A1 [5]
(b)	$f(x)=4x^3+x^2-11x+6=(4x-3)(x+2)(x-1)$	M1A1 [2]
(c)	$h(y)=2^{(3y+2)}+2^{2y}-11(2^y)+6 \Rightarrow h(y)=4(2^y)^3+(2^y)^2-11(2^y)+6$ $x=2^y \Rightarrow 2^y-1=0,\ \ 2^y+2=0,\ \ 4(2^y)-3=0$ $2^y=1 \Rightarrow y=0$ $2^y=\frac{3}{4} \Rightarrow y=\log_2\frac{3}{4}=-0.4150... \Rightarrow y=-0.415$ $[2^y=-2$ no solution$]$	M1 M1 B1 M1A1 [5]
		Total 12 marks

Question	Notes	Marks
(a)	For substituting ± 2 into $f(x)=0$ Allow sign errors	M1
	For substituting ± 1 into $f(x)=0$ Allow sign errors	M1
	For solving the resulting pair of simultaneous equations. $-13a+4+2b=0$ $5a+1-b=0$	M1
	For $a=2$ *	A1 cso
	For $b=11$	A1 [5]
	ALT uses polynomial division.	
	Multiplies out the product of the two factors to give: $(x+2)(x-1)=x^2+x-2$ This must be correct.	M1

Type MS · Two constants from factor and remainder information

Answer 13 of 13 · Question 5 part a · 5 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

	Uses polynomial division to give as a minimum: $2ax+(1-2M)$ where M is a constant. $x^2+x-2\overline{)2ax^3+x^2-bx+3a}$ $\quad 2ax+(1-2a)$	M1
	Equates coefficients: $(x^2+x-2)(2ax+(1-2a))=2ax^3+x^2-bx+3a$ $\Rightarrow 2ax^3+x^2+x(1-6a)+(-2+4a)=2ax^3+x^2-bx+3a$ $3a=4a-2$ $1-6a=-b$	M1
	For $a=2$ *	A1 cso
	For $b=11$	A1
(b)	Divides $f(x)=4x^3+x^2-11x+6$ by either $[x^2+x-2]$ or $x+2$ or $x-1$ Either, $\left[x^2+x-2\overline{)4x^3+x^2-11x+6}\right.$ $\quad 4x+k$ $x+2\overline{)4x^3+x^2-11x+6}$ $\quad 4x^2-7x\pm k$ $x-1\overline{)4x^3+x^2-11x+6}$ $\quad 4x^2+5x\pm k$	M1
	For the correct factorisation of $f(x)$ $4x^2-7x+3=(4x-3)(x-1)\Rightarrow f(x)=(4x-3)(x-1)(x+2)$	A1 [2]
(c)	For manipulating the indices to achieve $\Rightarrow h(y)=4(2^y)^3+(2^y)^2-11(2^y)+6$ OR Uses the factorised expression to obtain: $h(y)=(2^y+2)(2^y-1)(4(2^y)-3)$	M1
	Substitutes $x=2^y$ into the factorised f (x) to find either $2^y-1=0$ or $2^y+2=0$ or $4(2^y)-3=0$	M1
	For $2^y=1\Rightarrow y=0$	B1
	For $4(2^y)=3\Rightarrow 2^y=\frac{3}{4}\Rightarrow y=\log_2\frac{3}{4}$	M1
	For awrt $y=-0.4150...\Rightarrow y\approx -0.415$ [No solution for $2^y=-2$ - must reject]	A1 [5]
		Total 12 marks

Type MS · Two constants from factor and remainder information

Answer 13 of 13 · Question 5 part a · 5 marks

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Question	Scheme	Marks